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Root Capital and the Efficient Impact Frontier

Root Capital was founded in 1999 by Willy Foote as a nonprofit lender to agricultural enterprises in Africa and Latin America. Its mission was "to pioneer finance for grassroots business that build sustainable livelihoods and transform rural communities in poor, environmentally vulnerable areas." These businesses tended to fall in the "missing middle"—enterprises that were too large for micro-loans, but not yet deemed "bankable" by traditional, commercial lenders. **Exhibit 1** provides an overview of how Root Capital viewed the so called "missing middle." Root Capital's mission was not only to directly impact borrower enterprises and affiliated farmers, but also to create a bridge by which small enterprises could grow to the point that traditional lending sources would serve them.

Root Capital offered short-term working capital loans to finance enterprises' purchases of coffee, cocoa, and other crops from smallholder farmers, and long-term loans for capital expenditures such as new agro-processing facilities. Loans ranged from $50,000 to $2,000,000. From inception to the end of 2017, Root Capital had made more than 2,000 loans totaling $1.2 billion to more than 650 businesses, reaching 1.2 million farm families. Root Capital's average loan portfolio in 2017 was $86m, approximately 52% of which was in coffee, 34% in cocoa and other low-perishable export agricultural products, and 14% in other.

While repayment rates on most loans were high (94% on average), many loans were not in fact profitable. Root Capital intentionally sought out borrowers that could not obtain a loan from any other source, and geographies in which poverty and environmental degradation were severe. In some cases, borrowers could not obtain loans because they were located in remote or inaccessible areas. Other borrowers were simply too small to attract attention from commercial banks. Risk of default as well as underwriting costs of loans to these enterprises were high, while loan sizes (and hence interest margin to Root Capital) were low.

Root Capital therefore deliberately subsidized these loans, in part with philanthropic donations and in part with profits from other loans. In addition to its core lending activities, Root Capital provided financial advisory services to prospective clients to help them develop the skills they need to access and manage credit.

Root's overall business mix was, on average, deliberately concessionary, in order to achieve impact that it believed would not otherwise have occurred. Revenue from interest and fees covered approximately 60% of the costs of Root Capital's lending operation, while grants covered the

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remainder, as well as financial management training and impact assessment. Foundations provided 62% of this grant funding; corporate partners and government provided 15%; multilateral agencies each provided 15%; and individuals and NGOs provided the remainder. Root Capital raised debt at an average interest rate of 2.2% and a tenor of 3-12 years. Foundations provided 33% of this debt; government and multilateral agencies provided 24%; corporate partners and individuals each provided 18%; and commercial financial institutions provided 7%.

The fact that Root Capital sought to balance impact with financial risk and return created challenges both internally and externally. Mike McCreless (MBA, 2010), Senior Director of Strategy and Impact observed,

> Loan officers at Root Capital had developed intuitive ways of creating a portfolio to create both financial return and impact. Their intuition was a powerful tool—but intuition is fallible, and it isn't scalable. To achieve impact on a large scale and to do so efficiently, they needed analytical tools to manage risk, return, and impact in an integrated and quantitative way. . . . At the same time, it was incredibly hard for investors in impact funds such as Root Capital to tell the difference between good impact investing and 'impact-washing' on one hand, or simply bad financial investing on the other. The only way to determine the reality would be with data about the impacts and financial returns of impact investments, along with a framework for making sense that data—and at the time, neither existed.[a]

Starting in 2015, Root Capital began to develop tools to address these challenges.

Measuring and Managing Impact

Over the previous years, Root Capital had developed a robust, albeit partially siloed, approach to impact management. Loan officers screened each prospective borrower to either exclude those with socially and environmentally destructive practices, or to work with such borrowers to create a remediation plan, which would become part of the loan covenants. More importantly, during their onsite due diligence visits they also gathered extensive data about the positive social and environmental practices of all of its 300+ borrowers. In addition, Root Capital had conducted deeper impact studies with 31 enterprises, surveying more than 4,000 farmers and facilitating dozens of farmer focus groups on topics such as farm profitability, gender dynamics, and climate change.

Yet until 2015, much of this information remained in dedicated documents and templates maintained by Root Capital's small impact team. Impact data was reported to donors and investors, but loan officers did not have a way to actively prioritize loans based on their degree of expected positive impact. There was strong organization-wide commitment to Root Capital's mission in general terms, but there was not always clarity on which desired impacts were highest priority relative to others, nor on how these impacts should be weighed against financial risk and return. A scalable decision tool would require a more systematic and standardized approach.

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[a] McCreless, Michael. "Toward the Efficient Impact Frontier." *Stanford Social Innovation Review*, Winter 2017.

McCreless, Michael. "Bono Doesn't Know." *Stanford Social Innovation Review* online, April 7, 2017.

McCreless, Michael. "Constructing a portfolio on the efficient impact frontier within one asset class." Impact Management Project White Paper, October 2017.

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After much internal deliberation and experimentation, Root landed on a ten-point ex-ante rating of loans' expected impact. Loans were scored in two stages. In the first stage, Root Capital scored the expected impact of the loan on the enterprise, using the concept of additionality—the extent to which a given investment provided resources above and beyond what other investors would likely have provided. Low additionality (zero points) was assigned to any loan that likely would have been serviced by a commercial lender in Root Capital's absence. High additionality (6.5 points) was assigned to any loan which likely would not have been serviced by any other lender. An intermediate level of additionality (3.0 points) included loans that a borrower likely would have obtained from some other mission-driven organization, but not from a commercial lender.

In the second stage, Root Capital allocated up to 3.5 points for the expected impact of the enterprise on farmers and the environment. This included both the baseline levels of poverty and environmental vulnerability (0.5 points each), and the expected performance of the enterprise in addressing those baseline conditions (1.0 point each). Finally, Root Capital awarded 0.5 points for scale, as measured by the number of farmers and workers reached by an enterprise. See **Exhibit 2** for an overview of Root Capital's Expected Impact Rating.

The decision to emphasize or "overweight" additionality in the scoring was based on several factors. Most importantly, Root Capital believed that the loans they made which would not otherwise have been made generated significant impact. One quasi-experimental study,[b] for example, had demonstrated that on average, "less credit-worthy" enterprises in Root Capital's portfolio (35% of the total) were indeed credit-constrained, and that Root Capital's loans to these enterprises translated into their purchasing more product (33%) at higher prices (12%) from farmers then if the enterprise had not received the loan. Root believed this evidence indicated that, for high-additionality loans, farmers supplying the enterprise were better off (greater income) and that the enterprise itself was better off.[1]

Integrating Impact with Financial Risk and Return

The Expected Impact Rating, conducted by a loan officer on each potential loan and vetted by a member of the impact team, yielded a consistent measure of expected impact for all of Root Capital's loans. Root Capital's traditional underwriting processes already produced a financial "expected return," based on loan size and interest rate, underwriting costs, risk premium etc. In parallel to the expected impact rating, Root Capital developed a new financial risk rating for all of its loans, based on observed outcomes of more than 1,200 loans between 2005 and 2013. It conducted a cost analysis to estimate the operational costs incurred by specific types of loans. Finally, it created a new tool to estimate expected loan profitability based on all of these inputs. **Exhibit 3** provides an example calculation of the net profitability to Root Capital.

With these new impact and financial tools in place, Root Capital plotted its portfolio of loans on a graph with expected risk-adjusted financial return on the y-axis and expected impact on the x-axis. See **Figure 1**.

[b] Specifically, the study was a regression discontinuity design (RDD) comparing agricultural enterprises that received larger loans from Root Capital to essentially identical enterprises that received smaller loans. The difference in loan size was driven by being just above or just below one of the cut-offs used in the loan scoring system. As it is more or less arbitarary where the threshold is set and thus, which enterprises are below or above the line, this "approximates" a random experiment.

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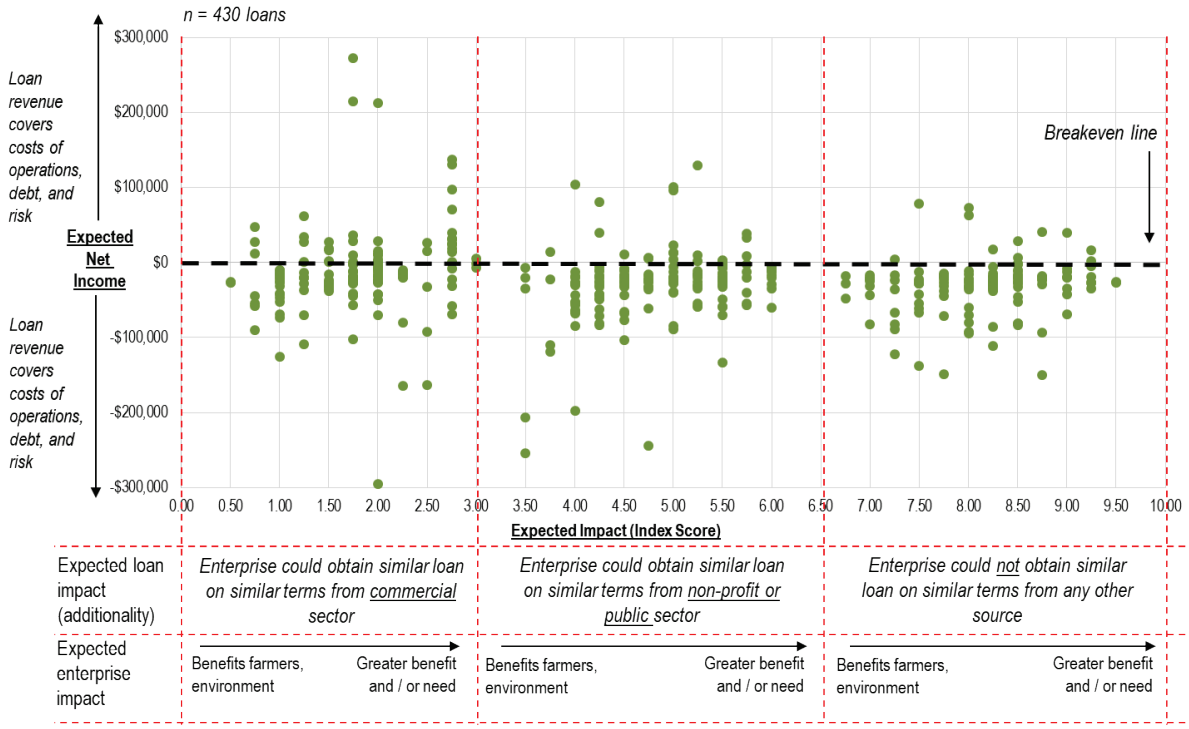
Figure 1 Expected Impact and Expected Risk-Adjusted Financial Return of Loans in Root Capital's Portfolio, 2016



Source: Company documents.

The Efficient Impact Frontier

Root Capital now had an integrated picture of financial risk/return and impact across its portfolio of loans. It could use this data to improve the impact and financial performance of its portfolio as a whole. Root Capital then developed the concept of an 'efficient impact frontier' (EIF) to support the next stage of portfolio construction decisions. The EIF was an extension of the concept of the 'efficient frontier' of risk and return, already well-known in finance. This was an aspirational set of portfolios, each of which would offer the greatest possible impact for any give financial return target. McCreless explained:

> Simply put, a portfolio that lies on the traditional efficient frontier offers the greatest possible financial return for a given level of risk and for a given set of investment opportunities. By extension, a portfolio of investments that lies on the 'efficient impact frontier' offers the highest level of overall impact, relative to the cumulative risk-adjusted financial return of those investments.

To illustrate this approach, Root Capital created a simulated efficient impact frontier for 2016 based on its existing portfolio in 2015. In particular, Root Capital conducted the following exercise to map out potentially better portfolios, taking as a point of departure the idea that they might have been able to allocate ten percent of their loan book differently. They did this in the following manner: first, they removed a randomly selected ten percent of loans from the 2015 portfolio. They then replaced these

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loans with the same number of loans, drawn from the original complete 2015 portfolio, without replacement. This exercise effectively assumed that Root Capital could have replicated any given loan one more time, but could not swap out more than 10% of its portfolio in a given year, and prevented identifying an unrealistic portfolio that cherry-picked only the very best loan and "replicated" it many times. This exercise gave Root Capital a sense of how much it might be able to improve its portfolio's impact, financial performance, or both. See **Exhibit 4** for sample portfolios.

While imperfect, this provided at least a theoretical framework by which to actively manage its loan book towards greater impact, given budget and return targets. Specifically, Root began to experiment with how it could inform loan decisions in 2017. "Up first, and then to the right" was the initial goal; although throughout the year, McCreless worked with the team to be able to articulate target portfolio goals for the next reporting period, with regards to both impact and financial performance, such as portfolio A, B or C in **Exhibit 4**.

Looking Forward

The organization had made significant progress since embarking on this new path in 2015, but many questions remained. What impact and financial targets should they set for the portfolio in 2017? How could those targets be informed by the actual set of investment opportunities that each loan officer observed for the coming year? Would Root Capital's donors and investors understand and accept the new approach?

Most importantly, would the approach enable Root Capital to direct more capital to where it was needed most, and could do the most to address poverty and environmental degradation? Might the approach even be adapted for use by other impact investors in other contexts?

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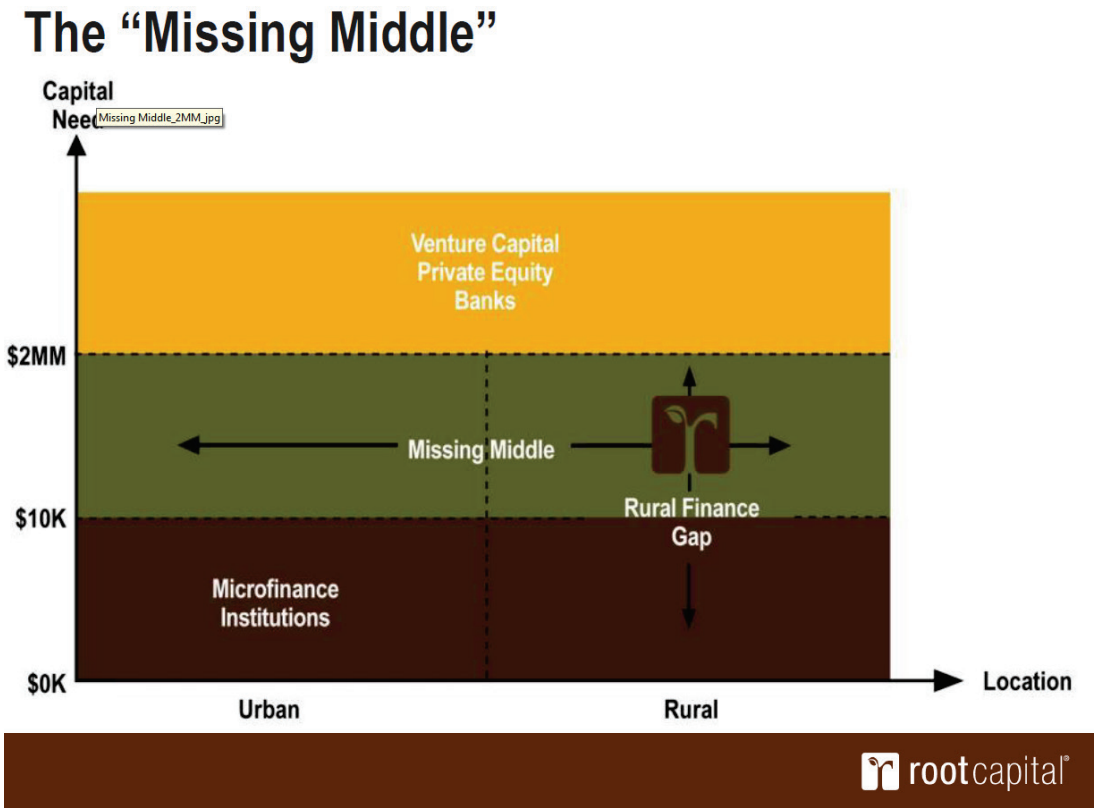
Exhibit 1 The Missing Middle



Source: Company documents.

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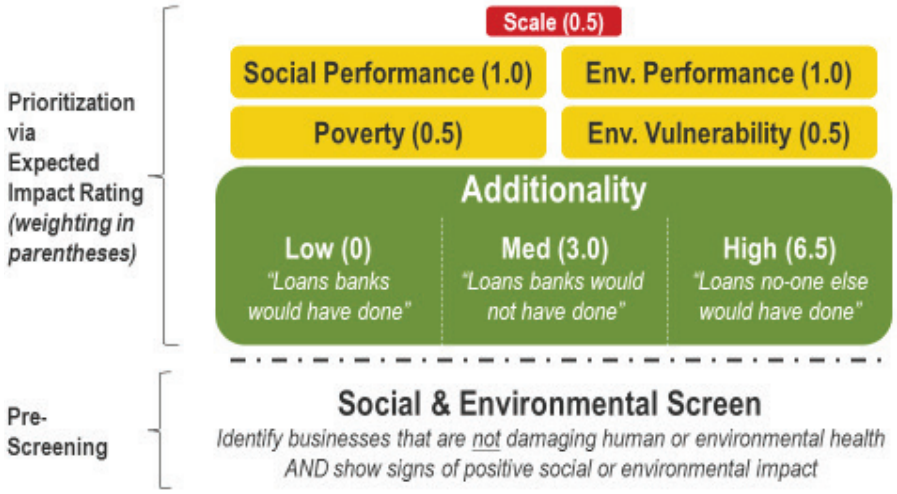
Exhibit 2 Root Capital's Expected Impact Rating



Source: Company documents.

Note: See **Appendix A** for more detail on the indicators, weights, and data sources in the Expected Impact Rating.

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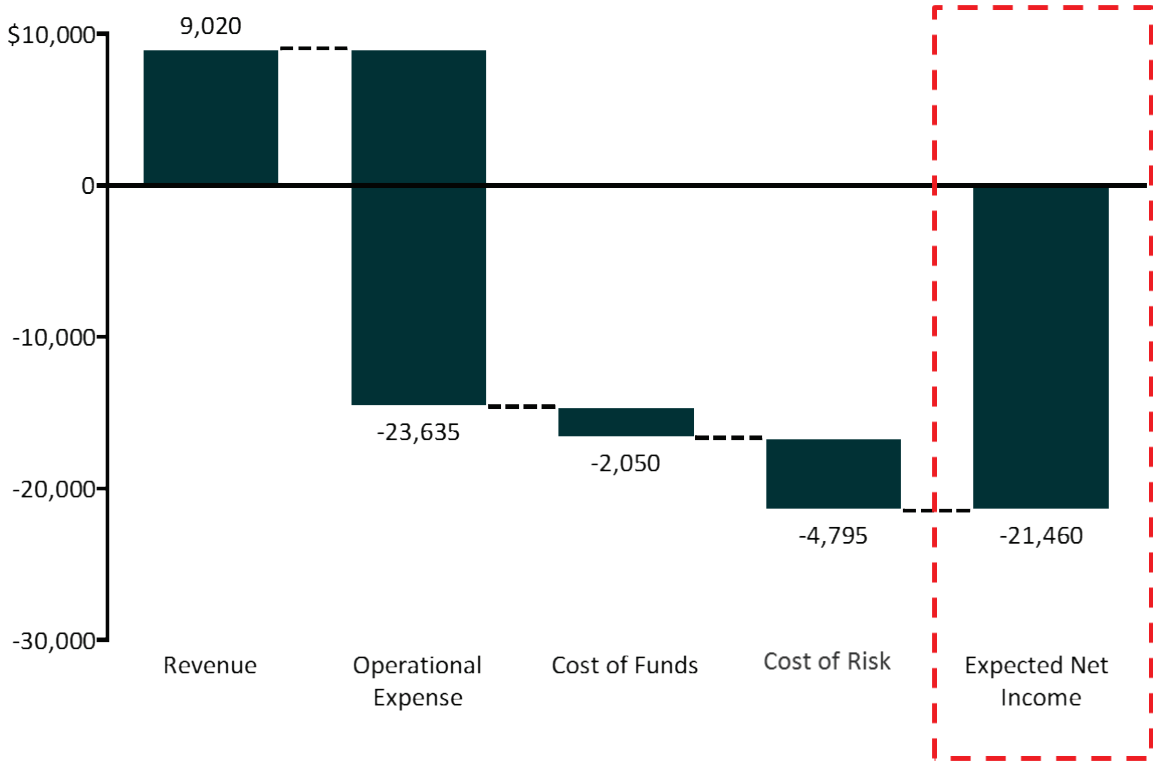
Exhibit 3 Expected Financial Return—Example Calculation



Source: Company documents.

Note: In this example, Root Capital makes an $87,000 loan to an early-stage coffee cooperative. In this example, the expected net profit is negative $21,000, because the cost of risk and particularly the operational costs outweigh the expected revenues. Root Capital covered half of this deficit with revenue from other, more profitable loans, and half with grant funding.

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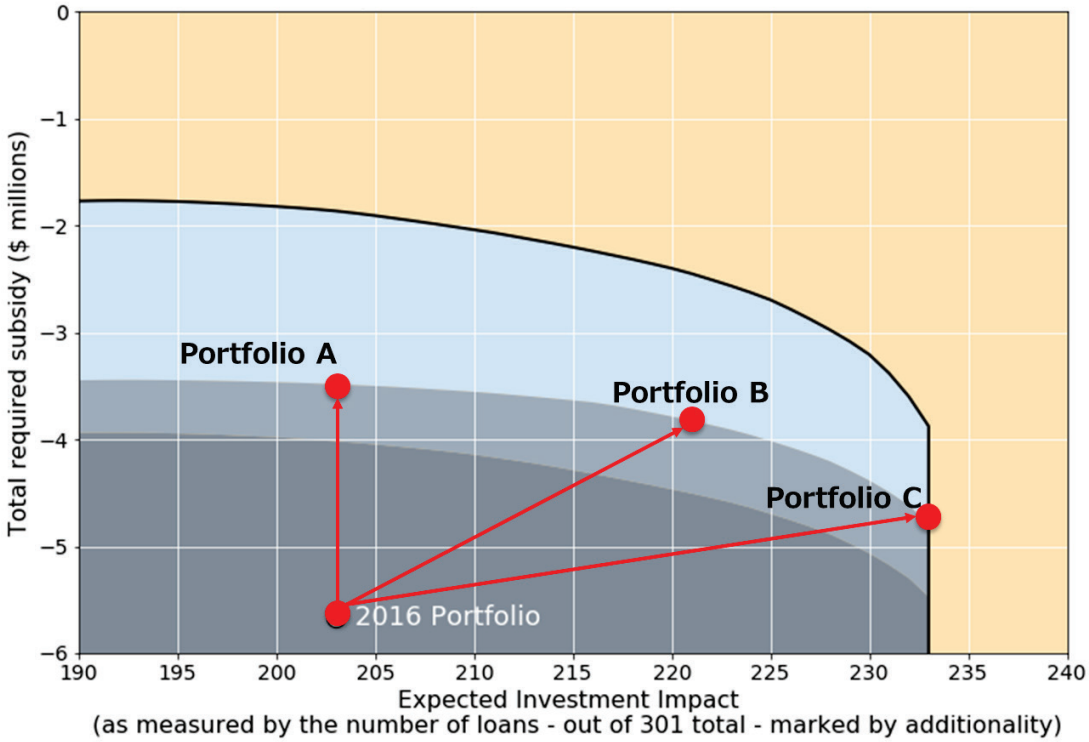
Exhibit 4 Illustrative Efficient Impact Frontier 2017



Source: Company documents.

Notes: Each point inside the shaded area represents a potential portfolio, and the line that marks the upper boundary of this area represents the efficient impact frontier. Each point on that line represents a simulated portfolio in which Root Capital eliminated the 10 percent of the sample loan portfolio that had the least impact and financial return, and replaced them with better-performing alternatives. For example, Portfolio A replaces the removed loans with duplicates of the highest-return loans in the current portfolio. Portfolio C replaces them with duplicates of the highest-impact loans (defined as loans scored highest in "additionality") in the current portfolio.

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Appendix A: Indicators, Weights, and Data Sources of Root Capital's Expected Impact Rating

Theme	Sub-Theme	Indicator	Points	Data Source
1. Expected Investment Impact / Additionality (up to 6.5 points)		Enterprise likely ***could not*** get a loan for this purpose, with similar collateral and for a rate & fee that is not more than 800 basis greater than Root Capital's, and in the same currency, from any other source than Root Capital[c]	6.5	Loan officer assessment based on discussion with enterprise managers and enterprise financials
		Enterprise likely **could** get a loan for this purpose, with similar collateral and for a rate & fee that is not more than 800 basis more than Root Capital's, and in the same currency, from ***a non-profit / public lender***	3.0	
		Enterprise likely ***could*** get a loan for this purpose, with similar collateral and for a rate & fee that is not more than 800 basis more than Root Capital's, and in the same currency, from a ***commercial lender***	0.0	
2. Expected Social Impact (up to 1.5 points)	Poverty Level (up to 0.5 point)	Extreme poverty country, or region within a country (< $2.50/day)	0.5	Progress out of Poverty (PPI) databases, Grameen Foundation.[2] To determine whether producers or employees affiliated with each enterprise are living below these poverty lines, we mapped PPI poverty rates to our clients' sourcing regions and areas of operation.
		High poverty country, or region within a country ($2.50-$4/person/day)	0.25	
		Moderate poverty country, or region within a country (>$/person/day)	0.0	
	Social Performance (up to 1.0 point)	• For farmers, paying at least 10% higher than local market price • For employees, paying more than 20% higher than the local minimum wage, or more than 10% higher plus health insurance/benefits • Providing on-farm agronomic training plus one of following: centralized training, access to inputs, access to on-farm equipment, both to over 50% of farmers • Providing or facilitating loans to over 25% of farmers • Providing a community service in education, health, access to water, or improved roads or transportation infrastructure to over 25% of farmers • Providing income diversification opportunities to over 25% of farmers	0.25 each; up to 1.0 max	Enterprise records; if enterprise is certified (e.g., organic or Fair Trade), records of certification audit; loan officer discussion with enterprise managers; spot checks of enterprise operations and discussions with affiliated farmers and employees.

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Theme	Sub-Theme	Indicator	Points	Data Source
		• Food security & nutrition focus: selling food into local markets to improve food security • Gender inclusive: 30% or more participation by women as producers and/or employees; OR women-led ***and*** 20% or more participation by women		
3. Expected Environmental Impact (up to 1.5 points)	Environmental Vulnerability[3] (up to 0.5 point)	Enterprise located in or sourcing from an "environmental degradation hotspot," defined as a region suffering from ongoing, significant degradation of one of the following natural resources: • Biodiversity: Region contains exceptional levels of native biodiversity and is currently threatened by significant degradation. • Soil: Region is experiencing significant downward pressure on chemical and/or physical components of soil health, with a rating of "degradation or very low [soil] resilience." • Water: Region with "extremely high risk" of water scarcity, based on its evaluation of local water quantity, water quality, and regulatory environment.	0.25	We used third-party, publically-available evaluations of environmental degradation for each natural resource: • Biodiversity: Critical Ecosystems Partnership Fund • Soil: Land Degradation Assessment in Drylands project of FAO, UNEP, and Global Environment Fund • Water: Aqueduct Water Risk Atlas, World Resources Institute To determine whether an enterprise operated in a particular hotspot, we created a map of enterprise operational and/or sourcing regions in Google Earth, and compared this map to maps created by the third-parties cited above.
		Enterprise located in or sourcing from a "climate change hotspot," in which climate change is expected to severely impact agricultural livelihoods. We classified a region as a climate hotspot if climate change probability maps indicated a 75%+ likelihood that climate change would push the area past either of two bioclimatic thresholds: • Maximum annual temperatures would flip above a key tolerance threshold for crops (30C); or • Length of the crop growing period would decline by 5%+.	0.25	We used analysis and maps produced by the CGIAR Research Program on Climate Change, Agriculture, and Food Security: Ericksen P, Thornton P, Notenbaert A, Cramer L, Jones P, Herrero M. 2011. Mapping hotspots of climate change and food insecurity in the global tropics. CCAFS Report no. 5. CGIAR Research Program on Climate Change, Agriculture and Food Security (CCAFS). Copen-hagen, Denmark. Available online at: www.ccafs.cgiar.org. To determine whether an enterprise operated in a climate change hotspot, we created a map of enterprise operational and/or sourcing regions in Google Earth, and compared this map to the CGIAR probability maps.

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Theme	Sub-Theme	Indicator	Points	Data Source
	Environmental Performance (up to 1.0 point)	Enterprise has an active environmental certification, including Aquaculture Stewardship Council, Fair trade (Fair Trade USA, FLO, IMO), Fair Wild, Forest Stewardship Council, Marine Stewardship Council, Organic (IFOAM standards only), Rainforest Alliance, Roundtable on Sustainable Palm Oil, Smithsonian Migratory Bird Council, Utz	0.5	Enterprise and certifier records
		Enterprise is engaged in climate change mitigation and / or adaptation activities, including: • Enterprise manages or sources from farmers managing ***diversified agroforestry*** (i.e., traditional polycultural or rustic) agricultural systems • Enterprise manages or sources from ***wild tree stands*** (i.e., traditionally-managed Brazil Nut or Shea) • Enterprise engages in or facilitates farmer engagement in ***a/reforestation***, resulting in the planting of > 100 trees / year • Enterprise uses or distributes ***clean or appropriate technologies***, such as solar panels or biodigestors, that deliver recognized reductions in greenhouse gas emissions compared to industry-standard alternatives • Enterprise provides farmers with weather/crop early warning system • Enterprise engages in another form of adaptation or mitigation work, either alone or with partners	0.25 each, up to 0.5	Enterprise records and partner records as relevant
4. Scale (up to 0.5 point)		Enterprise reaches <500 farmers and employees	0	Enterprise records
		Enterprise reaches 500–1,500 farmers and employees	0.25	
		Enterprise reaches > 1,500 farmers and employees	0.5	

Source: Company documents.

[c] The threshold of 800 basis points was chosen to be conservative in counting our additionality. For instance, if we are offering a loan with an interest rate of 11% and a commercial bank is offering a loan with a rate of 19%, we would count that loan as having zero additionality.

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Endnotes

[1] Macchiavello, Rocco and Arthur Blouin, "Strategic Default in the International Coffee Markets," Working Paper August 2017, reference number F-89305-CCP-1.

[2] Root Capital selected Grameen Foundation's Progress out of Poverty Index because it offered the best combination of comparability across countries and specificity at the region or district level within countries. The Progress out of Poverty Index estimates the likelihood that households in a defined geographic area are living on less than $2.50 / person / day. If the likelihood is greater than 50%, Root assumes that the average household in that area lives on less than $2.50 / person / day. If the likelihood is less than 50%, Root assumes that the average household in the area lives on more than $2.50 / person / day. The PPI score gives no indication of the likelihood that the average household in a geographic area is living on more or less than $4 / person / day, so Root had to make an assumption. Specifically, Root coded geographic areas "more than $4 / day" if there is less than a 30% likelihood that households in that area are living on less than $2.50 / person / day.

[3] The vast majority of Root Capital clients operate in regions threatened by natural resource degradation or future risk due to climate change. Root Capital believed it was important that these businesses not contribute to the problem, and adopt environmentally sustainable practices. Root Capital therefore screened prospective borrowers during due diligence to identify such businesses. All else being equal, through the expected impact rating, Root Capital sought to prioritize helping these borrowers succeed over borrowers operating in less threatened ecosystems.

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